

Project Documentation Directory — Maritime Port Intelligence Platform (MPIP)

Welcome to the official documentation folder for the **Maritime Port Intelligence Platform (MPIP) / Maritime Port Enhanced**.

This repository documentation is designed to give students, developers, and academic/industry mentors a complete, 360-degree understanding of the project's architecture, data engineering pipelines, machine learning forecasting algorithms, frontend user interface, and operational workflows.

■ Project Overview at a Glance

- **Project Name:** Maritime Port Intelligence Platform (MPIP) / Maritime Port Enhanced
 - **One-Line Explanation:** An AI-powered, data-grounded maritime decision-support platform designed to deliver predictive cargo volume forecasting, trade lane momentum analytics, risk/anomaly detection, and interactive port scenario simulation.
 - **Main Objective:** Bridge fragmented maritime datasets and convert historical port operations into actionable, forward-looking decisions for port authorities, customs agencies, and shipping line operators.
 - **Primary Focus: Predictive Cargo Analytics & Intelligence** (specifically grounded on actual multi-year monthly port traffic records mapped to New Mangalore Port Authority — NMPA).
 - **Technologies:** Python 3.12, FastAPI, Streamlit, Pandas, NumPy, Holt-Winters Exponential Smoothing, Multi-Feature Ridge Regression, Plotly, Docker, REST APIs.
 - **Main Features:**
 1. **Predictive Cargo Analytics & Volume Forecasting** (Holt-Winters + Ridge Regression with 95% dynamic confidence bounds)
 2. **Commodity & Trade Demand Intelligence** (Trade lane momentum, YoY growth tracking, commodity price indices)
 3. **Cargo Anomaly & Risk Intelligence** (Isolation Forest, LSTM, Autoencoder & Transformer simulation with severity progression)
 4. **Explainable Forecast Drivers** (Quantified weights for hinterland consumer demand, monsoon swells, and market inputs)
 5. **Model Validation & Out-of-Sample Backtesting** (WAPE, MAPE, MAE, RMSE performance comparison vs Seasonal Naïve baseline)
 6. **Interactive What-If Cargo Scenario Simulator** (Stress-testing port throughput against vessel arrival spikes, trade demand shifts, and weather delays)
 7. **Executive Command Center & Digital Twin** (Real-time KPIs, berth utilization, live vessel positions, revenue trends, and operational stress-testing)
 8. **AI Maritime Copilot** (LangGraph 3-tier cognitive dispatcher with specialized agent roster)
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■ Recommended Documentation Reading Order

For maximum clarity when reviewing this project or preparing for a mentor presentation, follow this sequence:

Step	File Name	Purpose / What You Will Learn
1	<u>01 PROJECT OVERVIEW.md</u>	High-level summary, 30s / 1min / 3min mentor elevators pitches.
2	<u>02 PROBLEM STATEMENT.md</u>	Real-world port management challenges & cargo data fragmentation.
3	<u>03 SOLUTION EXPLANATION.md</u>	End-to-end processing pipeline from CSV data source to UI insight.
4	<u>04 SYSTEM ARCHITECTURE.md</u>	Full technical breakdown, ASCII architecture diagrams, module layers.
5	<u>05 COMPLETE PROJECT FLOW.md</u>	Step-by-step user interaction lifecycle and request execution flow.
6	<u>06 CARGO FOCUSED FLOW.md</u>	Detailed explanation of why and how Cargo is the central axis.
7	<u>07 FRONTEND EXPLANATION.md</u>	Streamlit dashboard design, UI components, state management & Plotly charts.
8	<u>08 BACKEND EXPLANATION.md</u>	FastAPI route handlers, service layer logic, response schemas.
9	<u>09 DATABASE AND DATA.md</u>	In-memory data store, CSV persistence, data ingestion & upload handler.
10	<u>10 AI ML EXPLANATION.md</u>	Time-series ML algorithms (Holt-Winters, Ridge, Naive baseline, WAPE metrics).
11	<u>11 API DOCUMENTATION.md</u>	Complete REST API endpoint reference table and example JSON payloads.
12	<u>12 DATASET EXPLANATION.md</u>	In-depth breakdown of CODE_PH_0 (2021-2026 NMPA cargo traffic).
13	<u>13 OLD VS NEW COMPARISON.md</u>	Structural comparison between generic maritime dashboard vs cargo-core version.
14	<u>14 CHANGES MADE.md</u>	Comprehensive log of major/minor enhancements, refactorings, and UI fixes.
15	<u>15 FEATURES IMPLEMENTED.md</u>	Feature checklist categorizing Implemented vs Partially Implemented items.
16	<u>16 TECH STACK.md</u>	Technology choice justification, library dependencies, deployment containers.
17	<u>17 MENTOR EXPLANATION.md</u>	Ready-to-speak mentor script with structured sections for oral evaluation.
18	<u>18 MENTOR QA.md</u>	30+ predicted mentor questions with basic, technical, and deep-dive answers.

19	<u>19 DEMO SCRIPT.md</u>	Step-by-step live demo walkthrough script (What to Show vs What to Say).
20	<u>20 LIMITATIONS AND FUTURE SCOPE.md</u>	Current architectural limits, data constraints, and realistic roadmap.

[!NOTE]

*All code references, filenames, function names, and dataset metrics referenced in these documentation files correspond to the active codebase in **CODE_PH_0**.*